Chemistry

Science Mock Practice - With Answers

126 questions | 9 modules

Answers and explanations reproduce the website-delivered key. They have been checked against the current source and its checking logic; they are not an independent scientific review.

Module 1: Properties of acids and bases

Chapter 7 · Acids and Bases | Chapter 7 · 7.1–7.8

1. Have bitter taste

    a) Acidic solutions

    b) Basic solutions

    c) Neutral solutions

    d) Amphoteric solutions

Website answer: b) Basic solutions

Explanation: Basic solutions have a bitter taste; acids taste sour.

2. A common property of acids and bases....

    a) Electrical conductivity

    b) Bitter taste

    c) Generate H^+ ions

    d) Generate OH^- ions

Website answer: a) Electrical conductivity

Explanation: Both acids and bases conduct electricity because they form ions in solution.

3. Acidic solutions turn litmus color

    a) Blue to red

    b) Blue to green

    c) Blue to yellow

    d) Red to blue

Website answer: a) Blue to red

Explanation: Acids turn blue litmus red.

4. Basic solutions turn litmus color

a) Blue to red

b) Red to green

c) Red to yellow

d) Red to blue

Website answer: d) Red to blue

Explanation: Bases turn red litmus blue.

5. A chemical that turns red litmus to blue

a) HCl

b) KCl

c) CH_3COOH

d) NaOH

Website answer: d) NaOH

Explanation: NaOH is a base, so it turns red litmus blue; HCl and CH_3COOH are acids and KCl is a salt.

6. Acidic solutions contain hydrogen ions …. Hydroxide ions

a) Half

b) Less than

c) Equal

d) More than

Website answer: d) More than

Explanation: Acidic solutions contain more hydrogen ions than hydroxide ions.

7. Solution with hydroxide ion concentration more than hydrogen ion concentration

a) Acidic solution

b) Neutral solution

c) Basic solution

d) Amphoteric solution

Website answer: c) Basic solution

Explanation: A solution with $[OH^-] > [H^+]$ is basic.

8. Neutral solution contains equal concentrations of hydrogen ion and ….. ions

a) Hydroxide

b) Oxygen

c) Chloride

d) Nitrogen

Website answer: a) Hydroxide

Explanation: A neutral solution has equal concentrations of H^+ and OH^-.

9. Which of the properties listed below is not characteristic of an acid?

a) a sour taste

b) a slippery feel

c) the ability to conduct an electric current

d) Reactivity with metals

Website answer: b) a slippery feel

Explanation: A slippery feel is a property of bases, not acids.

Module 2: Arrhenius and Brønsted-Lowry models

Chapter 7 · Acids and Bases | Chapter 7 · 7.9–7.22

10. Hydrogen ion with a covalent bond with water ...

a) H_3O^+

b) OH^-

c) H^+

d) H_2O

Website answer: a) H_3O^+

Explanation: H^+ bonds covalently to water to form the hydronium ion, H_3O^+.

11. Bonding of water with hydrogen ion gives...

a) Ammonium

b) Hydroxide

c) Hydronium

d) Hydrogen

Website answer: c) Hydronium

Explanation: Water + H^+ gives the hydronium ion, H_3O^+.

12. Arrhenius definition of a base is....

a) Produces H^+

b) Produces OH^-

c) Electron pair donor

d) Electron pair acceptor

Website answer: b) Produces OH^-

Explanation: An Arrhenius base produces OH^- ions in aqueous solution.

13. Which is a base according to Arrhenius theory?

a) CH_3COOH

b) HCl

c) H_2SO_4

d) $Mg(OH)_2$

Website answer: d) $Mg(OH)_2$

Explanation: $Mg(OH)_2$ releases OH^- ions, so it is an Arrhenius base.

14. Substance that accepts a proton is a base according to.....

a) Brønsted-Lowry

b) Ionic

c) Arrhenius

d) Lewis

Website answer: a) Brønsted-Lowry

Explanation: In the Brønsted-Lowry model, a base is a proton (H^+) acceptor.

15. Conjugate base of formic acid in the equation $HCOOH(aq) + H_2O(l) \rightleftharpoons HCOO^-(aq) + H_3O^+(aq)$

a) HCOOH

b) H_2O

c) $HCOO^-$

d) H_3O^+

Website answer: c) $HCOO^-$

Explanation: HCOOH loses H^+ to become $HCOO^-$, its conjugate base.

16. Conjugate base of HCN is....

a) HCN^-

b) CN^-

c) CN^{2-}

d) H_2CN

Website answer: b) CN^-

Explanation: $HCN - H^+ = CN^-$.

17. Self-ionization of water produces H^+ and OH^- ions where.....

a) More OH^- are there

b) Equal number

c) More H^+ are there

d) Less H^+ are there

Website answer: b) Equal number

Explanation: Self-ionization of water gives equal numbers of H^+ and OH^-.

18. Conjugate base of H_2SO_4 is...

a) HSO_4^+

b) H_2SO_3

c) HSO_4^-

d) HSO_2

Website answer: c) HSO_4^-

Explanation: $H_2SO_4 - H^+ = HSO_4^-$.

19. Conjugate base of H_3PO_4 is...

a) $H_3PO_4^-$

b) PO_4^{3-}

c) HPO_4^{2-}

d) $H_2PO_4^-$

Website answer: d) $H_2PO_4^-$

Explanation: $H_3PO_4 - H^+ = H_2PO_4^-$.

20. Conjugate acid of the base HCO_3^- is...

a) CO_3^{2-}

b) H_2CO_3

c) HCO_3

d) HCO_3^{2-}

Website answer: b) H_2CO_3

Explanation: $HCO_3^- + H^+ = H_2CO_3$.

21. Conjugate acid of the base NH_3 is...

a) NH_2

b) NH^-

c) NH_3^-

d) NH_4^+

Website answer: d) NH_4^+

Explanation: $NH_3 + H^+ = NH_4^+$, the ammonium ion.

22. According to Brønsted-Lowry, water is... in the reaction $NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$

a) Base

b) Acid

c) Conjugate acid

d) Conjugate base

Website answer: b) Acid

Explanation: Water donates H^+ to NH_3, so water acts as a Brønsted-Lowry acid.

23. The substance that acts as an acid and a base is...

a) Salt

b) Neutral

c) Buffer

d) Amphoteric

Website answer: d) Amphoteric

Explanation: An amphoteric substance can act as an acid and as a base.

24. A characteristic of an Arrhenius base…

a) Electrolyte

b) Tastes bitter

c) Dissociates to form OH^- ions

d) All of the above

Website answer: d) All of the above

Explanation: All three statements describe an Arrhenius base.

25. Which of the following is a Brønsted-Lowry Base?

a) HCl

b) H_3O^+

c) HCO_3^-

d) H_3PO_4

Website answer: c) HCO_3^-

Explanation: HCO_3^- can accept H^+, so it acts as a Brønsted-Lowry base.

26. What is the conjugate acid of the base CH_3NH_2?

a) $CH_3NH_3^+$

b) CH_3NH

c) NH_3

d) NH_4^+

Website answer: a) $CH_3NH_3^+$

Explanation: $CH_3NH_2 + H^+ = CH_3NH_3^+$.

27. Water is considered….

a) Amphoteric

b) Acid

c) Base

d) neutral

Website answer: a) Amphoteric

Explanation: Water is amphoteric.

Module 3: Polyprotic acids and Lewis acids

Chapter 7 · Acids and Bases | Chapter 7 · 7.23–7.31

28. Monoprotic acid gives ion…

a) One hydroxide

b) One nitrogen

c) One oxygen

d) One hydrogen

Website answer: d) One hydrogen

Explanation: A monoprotic acid gives one hydrogen ion per molecule.

29. HCl acid is….

a) Monoprotic

b) Diprotic

c) Triprotic

d) Tetraprotic

Website answer: a) Monoprotic

Explanation: HCl gives only one H^+, so it is monoprotic.

30. Polyprotic acid contains more than one….. ion

a) Oxygen

b) Nitrogen

c) Hydrogen

d) Fluoride

Website answer: c) Hydrogen

Explanation: A polyprotic acid can donate more than one hydrogen ion.

31. Diprotic acid is….

a) HCOOH

b) CH_3COOH

c) H_2SO_4

d) H_3PO_4

Website answer: c) H_2SO_4

Explanation: H_2SO_4 has two ionizable hydrogens: diprotic.

32. H_3PO_4 is of the type….

a) Monoprotic

b) Diprotic

c) Triprotic

d) Tetraprotic

Website answer: c) Triprotic

Explanation: H_3PO_4 can give three H^+: triprotic.

33. Acceptors of one pair of electrons are…

a) Lewis acids

b) Lewis bases

c) Brønsted-Lowry acids

d) Brønsted-Lowry bases

Website answer: a) Lewis acids

Explanation: A Lewis acid is an electron-pair acceptor.

34. Which is a Lewis acid?

a) PCl_3

b) BCl_3

c) H_2O

d) NH_3

Website answer: b) BCl_3

Explanation: Boron in BCl_3 has an incomplete octet, so it accepts an electron pair: Lewis acid.

35. Which is a Lewis acid?

a) F^-

b) Ag^+

c) Cl^-

d) NH_3

Website answer: b) Ag^+

Explanation: Ag^+ accepts electron pairs, so it is a Lewis acid; the others donate pairs.

36. PCl_3 is classified as ….. according to Lewis theory. Atomic number of P = 15

a) Base

b) Acid

c) Salt

d) Amphoteric

Website answer: a) Base

Explanation: PCl_3 has a lone pair it can donate, so it is a Lewis base.

37. What is the correct acid name for an aqueous solution of $HClO_4$?

a) Hypochlorous acid

b) Chlorous acid

c) Chloric acid

d) Perchloric acid

Website answer: d) Perchloric acid

Explanation: $HClO_4$ is perchloric acid.

38. How many protons per molecule can a diprotic acid donate

a) One

b) Two

c) Three

d) zero

Website answer: b) Two

Explanation: A diprotic acid can donate two protons per molecule.

39. Which substance is a Lewis acid in the reaction $BF_3(aq) + F^-(aq) \rightleftharpoons BF_4^-(aq)$

a) F^-

b) BF_4^-

c) BF_3

d) No Lewis acid

Website answer: c) BF_3

Explanation: BF_3 accepts an electron pair from F^-, so it is the Lewis acid.

Module 4: Strength, pH and pOH

Chapter 7 · Acids and Bases | Chapter 7 · 7.32–7.49

40. Which is a basic anhydride?

a) Carbon dioxide

b) Calcium oxide

c) Nitrogen dioxide

d) Sulfur oxide

Website answer: b) Calcium oxide

Explanation: Calcium oxide is a metal oxide that forms a base with water.

41. Which is a strong acid?

a) HCl

b) HF

c) H_2S

d) CH_3COOH

Website answer: a) HCl

Explanation: HCl is a strong acid; HF, H_2S and CH_3COOH are weak.

42. Which is a strong base?

a) SO_2

b) NH_3

c) NaCl

d) NaOH

Website answer: d) NaOH

Explanation: NaOH ionizes completely, so it is a strong base.

43. Product of hydrogen ion conc. and hydroxide ion conc. in dilute solutions is called...

a) Hydrogen number

b) Hydroxide number

c) Molarity

d) Water ionization constant

Website answer: d) Water ionization constant

Explanation: $[H^+][OH^-] = K_w$, the ion product (ionization) constant of water = 1×10^{-14}.

44. Which is true for an acidic solution?

a) $[H^+] = 10^{-9}$ M

b) $[H^+] = 10^{-14}$ M

c) $[H^+] > [OH^-]$

d) $[H^+] < [OH^-]$

Website answer: c) $[H^+] > [OH^-]$

Explanation: In an acidic solution $[H^+] > [OH^-]$.

45. If pH=2, then the solution…

a) About neutral

b) Acidic

c) Basic

d) pOH < 10

Website answer: b) Acidic

Explanation: pH 2 is below 7, so the solution is acidic.

46. The solution is basic if pH…

a) pH = 0

b) pH = 7

c) pH > 7

d) pH < 7

Website answer: c) pH > 7

Explanation: A solution is basic when pH > 7.

47. Which is a hydrogen number of a base?

a) 8

b) 6

c) 2

d) 1

Website answer: a) 8

Explanation: 8 is the only value above 7.

48. If the hydronium ion conc. is 10^{-9}M, then the solution is…

a) Neutral

b) Basic

c) Amphoteric

d) Acidic

Website answer: b) Basic

Explanation: $[H_3O^+] = 10^{-9}$ M gives pH 9, which is basic.

49. pH of a strong acid…

a) 14

b) 7

c) 4

d) 1

Website answer: d) 1

Explanation: A strong acid has a pH close to 1.

50. If pH = 1.6, the solution…

a) Weak acid

b) Strong acid

c) Weak base

d) Strong base

Website answer: b) Strong acid

Explanation: pH 1.6 is far below 7: a strong acid.

51. If $[H^+] = 1\times10^{-13}$, the solution is…

a) Strong acid

b) Weak acid

c) Strong base

d) Weak base

Website answer: c) Strong base

Explanation: pH = 13, which is a strong base.

52. pOH of a strong base…

a) 14

b) 7

c) >7

d) 0

Website answer: d) 0

Explanation: A strong base has a very low pOH, close to 0.

53. If pH=5.2, then pOH equals…

a) 14

b) 5.2

c) 8.8

d) 0

Website answer: c) 8.8

Explanation: pOH = 14 − 5.2 = 8.8.

54. If pOH = 3, then $[H^+]$ equals…

a) 1×10^{-11} M

b) 1×10^{-8} M

c) 11 M

d) 3 M

Website answer: a) 1×10^{-11} M

Explanation: pH = 14 − 3 = 11, so $[H^+] = 1 \times 10^{-11}$ M.

55. Find pH if $[OH^-] = 10^{-5}$ M

a) 9

b) 5

c) 4

d) 2

Website answer: a) 9

Explanation: pOH = 5, so pH = 14 − 5 = 9.

56. Which is used to estimate hydrogen number?

a) Hydrometer

b) Manometer

c) Litmus paper

d) Venturi measurement

Website answer: c) Litmus paper

Explanation: Litmus paper (an indicator) estimates whether the pH is acidic or basic.

57. Hydrogen number can be measured by…

a) Digital pH meter

b) Hydrometer

c) Manometer

d) Venturi measurement

Website answer: a) Digital pH meter

Explanation: A digital pH meter measures the pH accurately.

58. Which is an example of a weak acid?

a) HCl

b) HNO_3

c) H_2SO_4

d) CH_3COOH

Website answer: d) CH_3COOH

Explanation: Acetic acid (CH_3COOH) is a weak acid.

Module 5: Neutralization, titration, salts and buffers

Chapter 7 · Acids and Bases | Chapter 7 · 7.50–7.62

59. Acid + base → salt + water. This is …reaction

a) Combustion

b) Decomposition

c) Single displacement

d) Double displacement

Website answer: d) Double displacement

Explanation: Acid + base → salt + water is a double displacement (neutralization) reaction.

60. An ionic compound composed of positive ion from the base and negative ion from the acid…

a) Base

b) Acid

c) Salt

d) Water

Website answer: c) Salt

Explanation: A salt is an ionic compound of the cation from the base and the anion from the acid.

61. The use of acid or base to know the conc. of the other is…

a) Titration

b) Combustion

c) Distillation

d) Hydrolysis

Website answer: a) Titration

Explanation: Titration uses an acid or base of known concentration to find the concentration of the other.

62. Solution with known conc. used in titration…

a) Standard solution

b) Buffer solution

c) Conc. solution

d) Diluted solution

Website answer: a) Standard solution

Explanation: A standard solution has a known concentration.

63. At equivalence point, moles of H^+...moles of OH^-

a) More than

b) Equal

c) Less than

d) Double

Website answer: b) Equal

Explanation: At the equivalence point, moles of H^+ = moles of OH^-.

64. Chemical dyes affected in color in acid or base...

a) Mixtures

b) Indicators

c) Standard solutions

d) Buffer

Website answer: b) Indicators

Explanation: Indicators are dyes whose colour depends on acidity or basicity.

65. At the end of titration, ... will change color

a) Indicator

b) Acid

c) Base

d) Salt

Website answer: a) Indicator

Explanation: The indicator changes colour at the end point of the titration.

66. In hydrolysis of salts, negative ion will gain...

a) Hydrogen ions

b) Hydroxide ions

c) Nitrogen ions

d) Oxygen ions

Website answer: a) Hydrogen ions

Explanation: The negative ion gains hydrogen ions (from water), leaving OH^-.

67. Which is an acidic salt?

a) Potassium fluoride

b) Ammonium chloride

c) Sodium nitrate

d) Lithium hydroxide

Website answer: b) Ammonium chloride

Explanation: Ammonium chloride (from HCl and NH_3) is an acidic salt.

68. Resists a change in its pH when H^+ or OH^- ions are added.

a) Titrant solution

b) Buffer

c) Standard solution

d) Conc. solution

Website answer: b) Buffer

Explanation: A buffer resists changes in pH when H^+ or OH^- are added.

69. Produced by adding weak base to its conjugate acid…

a) Standard solution

b) Diluted solution

c) Saturated solution

d) Buffer

Website answer: d) Buffer

Explanation: A weak base mixed with its conjugate acid forms a buffer.

70. Adding acid to a buffer… the value of pH

a) Not change

b) Increase

c) Decrease

d) Double

Website answer: c) Decrease

Explanation: Adding acid to a buffer slightly decreases the pH.

71. Buffer capacity…molecules and ionic conc. in the solution

a) Increase by decrease

b) Increase by increase

c) No change by increase

d) No change by decrease

Website answer: b) Increase by increase

Explanation: Buffer capacity increases as the concentrations of the molecules and ions increase.

72. Neutralization of any strong acid and strong base produces mostly

a) H_2O molecules

b) H_3O^+ and OH^- ions

c) H_3O^+ ions

d) OH^- ions

Website answer: a) H_2O molecules

Explanation: Neutralization of a strong acid and strong base produces mostly water molecules.

Module 6: Oxidation, reduction and redox agents

Chapter 8 · Electrochemistry | Chapter 8 · 8.1–8.10

73. In oxidation the oxidation number…

a) Equals zero

b) Doesn't change

c) Decreases

d) Increases

Website answer: d) Increases

Explanation: In oxidation the oxidation number increases.

74. Which is an oxidation process?

a) $Ag(s) \rightarrow Ag^{+}(aq) + e^{-}$

b) $Hg^{2+}(aq) + 2e^{-} \rightarrow Hg(l)$

c) $Al^{3+}(aq) + 3e^{-} \rightarrow Al(s)$

d) $Mg^{2+}(aq) + 2e^{-} \rightarrow Mg(s)$

Website answer: a) $Ag(s) \rightarrow Ag^{+}(aq) + e^{-}$

Explanation: $Ag \rightarrow Ag^{+} + e^{-}$ loses an electron: oxidation.

75. In the equation $MnO_2 + Fe^{3+} \rightarrow MnO_4^{-} + Fe^{2+}$

a) Manganese oxidized

b) Iron oxidized

c) Manganese reduced

d) Oxygen oxidized

Website answer: a) Manganese oxidized

Explanation: The oxidation number of Mn increases, so manganese is oxidized.

76. Which is the reducing agent in the reaction $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$

a) $Zn^{2+}(aq)$

b) $Cu(s)$

c) $Zn(s)$

d) $Cu^{2+}(aq)$

Website answer: c) $Zn(s)$

Explanation: $Zn(s)$ loses electrons (is oxidized), so it is the reducing agent.

77. In the half reaction $Fe(s) \rightarrow Fe^{2+}(aq) + 2e^-$

a) Iron is reducing agent

b) Iron gained $2e^-$

c) Iron is oxidizing agent

d) Ox. number decreases

Website answer: a) Iron is reducing agent

Explanation: Iron is oxidized, so it is the reducing agent.

78. Which is correct for the reaction $2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$

a) Na is reducing agent

b) Cl_2 ox.no increased

c) Na is oxidizing agent

d) Cl_2 is reducing agent

Website answer: a) Na is reducing agent

Explanation: Na is oxidized, so it is the reducing agent.

79. The reducing agent in the reaction $H_2S(g) + Cl_2(g) \rightarrow S(s) + 2HCl(g)$

a) $H_2(g)$

b) $Cl_2(g)$

c) $H_2S(g)$

d) $HCl(g)$

Website answer: c) $H_2S(g)$

Explanation: H_2S is oxidized, so it is the reducing agent.

80. An element is strong oxidizing agent if....

a) Octet

b) High electronegative

c) Low ionization energy

d) High boiling point

Website answer: b) High electronegative

Explanation: High electronegativity makes an element a strong oxidizing agent.

81. In the reaction $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$

a) Zn(s) oxidizing agent

b) $Cu^{2+}(aq)$ reducing agent

c) $Cu^{2+}(aq)$ oxidizing agent

d) Zn(s) reduced

Website answer: c) $Cu^{2+}(aq)$ oxidizing agent

Explanation: Cu^{2+} is reduced, so it is the oxidizing agent.

82. Element with valency +2 is....

a) Nonmetal

b) Metal

c) Semimetal

d) Inert

Website answer: b) Metal

Explanation: An element with valency +2 is a metal (e.g. Mg, Ca).

83. During oxidation, one or more electrons are removed from a substance, which increases its...

a) Size

b) Mass

c) Valence

d) Oxidation number

Website answer: d) Oxidation number

Explanation: Removing electrons increases the oxidation number.

84. During a redox reaction, the oxidation numbers of oxidizing agents....

a) Decrease

b) Increase

c) Remain the same

d) double

Website answer: a) Decrease

Explanation: Oxidizing agents gain electrons, so their oxidation numbers decrease.

Module 7: Oxidation numbers

Chapter 8 · Electrochemistry | Chapter 8 · 8.11–8.19

85. Oxidation number of Al in its compounds is…

a) -3

b) +2

c) -1

d) +3

Website answer: d) +3

Explanation: Aluminum has oxidation number +3 in its compounds.

86. Valency of $_{10}$Ne is

a) -10

b) -5

c) 0

d) 10

Website answer: c) 0

Explanation: Neon is inert; its valency is 0.

87. Oxidation number of oxygen in H_2O_2 is…

a) 0

b) +1

c) -1

d) +2

Website answer: c) -1

Explanation: In H_2O_2 oxygen has oxidation number -1.

88. Oxidation number of sulfur in H_2SO_4 …

a) -2

b) +6

c) +2

d) +3

Website answer: b) +6

Explanation: S = +6 in H_2SO_4.

89. Oxidation number of nitrogen in HNO_2

a) -2

b) +2

c) +5

d) +3

Website answer: d) +3

Explanation: N = +3 in HNO_2.

90. Oxidation number of nitrogen in N_2O_4

a) +2

b) +4

c) -2

d) -4

Website answer: b) +4

Explanation: N = +4 in N_2O_4.

91. Oxidation number of iron in FeO

a) +3

b) -3

c) +2

d) -2

Website answer: c) +2

Explanation: Fe = +2 in FeO.

92. Oxidation number of iron in $Fe(OH)_3$

a) +1

b) -1

c) -3

d) +3

Website answer: d) +3

Explanation: Fe = +3 in $Fe(OH)_3$.

93. Oxidation number of Cr in $K_2Cr_2O_7$

a) -12

b) +12

c) -6

d) +6

Website answer: d) +6

Explanation: Cr = +6 in $K_2Cr_2O_7$.

94. What is the oxidation number of Mn in MnO_4^-?

a) +4

b) +5

c) +6

d) +7

Website answer: d) +7

Explanation: Mn = +7 in MnO_4^-.

95. What is the half-reaction of Zn in the reaction Zn(s) + HgO → ZnO + Hg(s)?

a) $Zn(s) \rightarrow Zn^{+1}$

b) $Zn(s) \rightarrow Zn^{+2} + 2e^-$

c) $Zn(s) \rightarrow Zn^{+1} + 2e^-$

d) $Zn(s) \rightarrow Zn^{+2} + 1e^-$

Website answer: b) $Zn(s) \rightarrow Zn^{+2} + 2e^-$

Explanation: $Zn(s) \rightarrow Zn^{2+} + 2e^-$.

Module 8: Galvanic cells and cell potential

Chapter 8 · Electrochemistry | Chapter 8 · 8.20–8.33

96. Study of change of chemical energy to electrical energy by redox reactions....

a) Analytical chemistry

b) Atomic chemistry

c) Biochemistry

d) Electrochemistry

Website answer: d) Electrochemistry

Explanation: Electrochemistry studies converting chemical energy to electrical energy by redox reactions.

97. Anode is the electrode where .....takes place

a) Reduction

b) Oxidation

c) Gain of electrons

d) Precipitation

Website answer: b) Oxidation

Explanation: Oxidation takes place at the anode.

98. Galvanic cell is a type of....cell

a) Electromagnetic

b) Electrochemical

c) Electro thermal

d) Chemical

Website answer: b) Electrochemical

Explanation: A galvanic cell is a type of electrochemical cell.

99. Electric current is produced from chemical reactions by...

a) Metal corrosion

b) Electrolysis

c) Electroplating

d) Galvanic cells

Website answer: d) Galvanic cells

Explanation: Galvanic cells produce electric current from chemical reactions.

100. Which is a correct half reaction equation?

a) $Ag^{+}(aq) \rightarrow Ag(s) + e^{-}$

b) $Hg^{2+}(aq) \rightarrow Hg(l) + 2e^{-}$

c) $2O^{2-}(aq) + 2e^{-} \rightarrow O_2(g)$

d) $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^{-}$

Website answer: d) $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^{-}$

Explanation: $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^{-}$ is a correct, balanced oxidation half-reaction.

101. Ions move in the galvanic cell through…

a) Cathode

b) Anode

c) Wire

d) Salt bridge

Website answer: d) Salt bridge

Explanation: Ions move through the salt bridge.

102. A tube with KCl solution allows ions to move ..

a) Cathode

b) Anode

c) Salt bridge

d) Hydrogen electrode

Website answer: c) Salt bridge

Explanation: The salt bridge is a tube with an electrolyte that lets ions move.

103. Energy that pushes electrons from anode to cathode

a) Potential elec. energy

b) Cathode potential

c) Anode potential

d) Potential difference

Website answer: d) Potential difference

Explanation: The potential difference pushes electrons from the anode to the cathode.

104. Reduction potential is the ability to…

a) Decompose

b) Gain electrons

c) Lose electrons

d) Oxidation

Website answer: b) Gain electrons

Explanation: Reduction potential is the tendency to gain electrons.

105. Standard reduction potential equals...

a) 0 V

b) 1 V

c) -1 V

d) -1.1 V

Website answer: a) 0 V

Explanation: The standard hydrogen electrode's reduction potential is defined as 0 V.

106. Which is the cell potential equation?

a) E°cell = E°cathode + E°anode

b) E°cell = E°anode - E°cathode

c) E°cell = E°anode + E°cathode

d) E°cell = E°cathode - E°anode

Website answer: d) E°cell = E°cathode - E°anode

Explanation: E°cell = E°cathode − E°anode.

107. Find the cell potential for $Sn(s) + Cu^{2+}(aq) \rightarrow Sn^{2+}(aq) + Cu(s)$, $E°Sn^{2+} = -0.1$ V, $E°Cu^{2+} = +0.3$V

a) 0.1 V

b) 0.2 V

c) 0.3 V

d) 0.4 V

Website answer: d) 0.4 V

Explanation: $E°cell = 0.3 - (-0.1) = 0.4$ V.

108. In spontaneous reaction, the cell potential is...

a) Negative

b) Positive

c) High

d) Low

Website answer: b) Positive

Explanation: A spontaneous cell reaction has a positive cell potential.

109. In the cell $Mg(s) + Pb^{2+}(aq) \rightarrow Pb(s) + Mg^{2+}(aq)$, $E°Pb^{2+} = -0.126V$, $E°Mg^{2+} = -2.37V$. Is it....

a) Spontaneous

b) Nonspontaneous

c) Reversible

d) Incomplete

Website answer: a) Spontaneous

Explanation: E°cell = +2.244 V, positive, so the reaction is spontaneous.

110. The following pair of half-reactions occurs in an electrochemical cell: $Ag^{+} + e^{-} \rightarrow Ag$, $H_2S \rightarrow S + 2H^{+} + 2e^{-}$. Where does the silver half-reaction take place?

a) At the anode

b) At the cathode

c) In salt bridge

d) On the wire

Website answer: b) At the cathode

Explanation: Reduction happens at the cathode.

111. What is the Ecell for the cell: $Zn^{2+} + 2e^{-} \rightleftharpoons Zn$ ($E° = -0.76$ V), $Hg^{2+} + 2e^{-} \rightleftharpoons Hg$ ($E° = +0.85$ V)

a) +1.56 V

b) -1.56 V

c) +0.04 V

d) -0.04 V

Website answer: a) +1.56 V

Explanation: E°cell = 0.85 − (−0.76) = +1.61 V; the closest choice is +1.56 V, and it is positive.

112. Chemical energy is converted into electrical energy in a(n)

a) Half cell

b) Galvanic cell

c) Electrolytic cell

d) Electroplating cell

Website answer: b) Galvanic cell

Explanation: A galvanic cell converts chemical energy into electrical energy.

Module 9: Batteries, corrosion and electrolysis

Chapter 8 · Electrochemistry | Chapter 8 · 8.34–8.44

113. In zinc carbon battery, the cathode is…

a) Carbon

b) Zinc

c) Copper coil

d) KOH

Website answer: a) Carbon

Explanation: In a zinc-carbon battery, the carbon rod is the cathode.

114. Zinc powder mixed with KOH is the anode in…

a) Lithium battery

b) Silver battery

c) Galvanic cell

d) Lead battery

Website answer: b) Silver battery

Explanation: In a silver battery, zinc powder mixed with KOH is the anode.

115. What is the source of energy in earphone?

a) Dry cell

b) Alkaline cell

c) Silver battery

d) Lead battery

Website answer: a) Dry cell

Explanation: The book's answer is the dry cell.

116. Production of electricity by reversible redox reaction…

a) Alkaline battery

b) Dry battery

c) Secondary battery

d) Silver battery

Website answer: c) Secondary battery

Explanation: Secondary (rechargeable) batteries depend on reversible redox reactions.

117. Depends on reversible redox reaction…

a) Alkaline battery
b) Silver battery
c) Dry battery
d) Secondary battery

Website answer: d) Secondary battery

Explanation: A secondary battery depends on reversible redox reactions.

118. Example of a secondary battery is

a) Alkaline battery
b) Silver battery
c) Laptop battery
d) Zn/C battery

Website answer: c) Laptop battery

Explanation: A laptop battery is rechargeable, so it is a secondary battery.

119. Loss of metal by redox reactions with surroundings is…

a) Ionization
b) Galvanization
c) Corrosion
d) decomposition

Website answer: c) Corrosion

Explanation: Corrosion is the loss of metal by redox reactions with the surroundings.

120. Coating of iron with a high corrosion resistant…

a) Decomposition
b) Galvanization
c) Ionization
d) Precipitation

Website answer: b) Galvanization

Explanation: Galvanization coats iron with a corrosion-resistant metal such as zinc.

121. Separation of water components by…

a) Filtration
b) Crystallization
c) Sublimation
d) Electrolysis

Website answer: d) Electrolysis

Explanation: Electrolysis separates water into hydrogen and oxygen.

122. To obtain chlorine from electrolysis of molten NaCl we use…

a) Downs cell

b) Galvanization

c) Hall-Héroult process

d) Halogenation

Website answer: a) Downs cell

Explanation: The Downs cell produces chlorine and sodium from molten NaCl.

123. The method to produce hydrogen, aluminum, and metal purification….

a) Electrolysis

b) Galvanization

c) Electroplating

d) Thermal cracking

Website answer: a) Electrolysis

Explanation: Electrolysis is used to produce hydrogen and aluminum and to purify metals.

124. The cathode in a zinc-carbon dry cell is

a) Auto-oxidizing

b) Carbon electrolyte

c) Carbon electrode

d) Zinc electrode

Website answer: c) Carbon electrode

Explanation: The cathode in a zinc-carbon dry cell is the carbon electrode.

125. Which process breaks down a compound into its elements?

a) Electrolysis

b) Electroplating

c) Corrosion

d) Reduction

Website answer: a) Electrolysis

Explanation: Electrolysis breaks a compound into its elements.

126. In an electroplating cell, the object to be plated is the

a) Cathode

b) Anode

c) Electrolyte

d) Salt bridge

Website answer: a) Cathode

Explanation: The object to be plated is the cathode.

Complete answer key

Question numbers match the question bank. Letters refer to the original choice order.

1. b)	2. a)	3. a)	4. d)	5. d)	6. d)
7. c)	8. a)	9. b)	10. a)	11. c)	12. b)
13. d)	14. a)	15. c)	16. b)	17. b)	18. c)
19. d)	20. b)	21. d)	22. b)	23. d)	24. d)
25. c)	26. a)	27. a)	28. d)	29. a)	30. c)
31. c)	32. c)	33. a)	34. b)	35. b)	36. a)
37. d)	38. b)	39. c)	40. b)	41. a)	42. d)
43. d)	44. c)	45. b)	46. c)	47. a)	48. b)
49. d)	50. b)	51. c)	52. d)	53. c)	54. a)
55. a)	56. c)	57. a)	58. d)	59. d)	60. c)
61. a)	62. a)	63. b)	64. b)	65. a)	66. a)
67. b)	68. b)	69. d)	70. c)	71. b)	72. a)
73. d)	74. a)	75. a)	76. c)	77. a)	78. a)
79. c)	80. b)	81. c)	82. b)	83. d)	84. a)
85. d)	86. c)	87. c)	88. b)	89. d)	90. b)
91. c)	92. d)	93. d)	94. d)	95. b)	96. d)
97. b)	98. b)	99. d)	100. d)	101. d)	102. c)
103. d)	104. b)	105. a)	106. d)	107. d)	108. b)
109. a)	110. b)	111. a)	112. b)	113. a)	114. b)
115. a)	116. c)	117. d)	118. c)	119. c)	120. b)
121. d)	122. a)	123. a)	124. c)	125. a)	126. a)